

Privacy with Asymmetrically Informed Consumers

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Abstract

I study privacy when consumers are uncertain about what their data will be used for. I begin by reviewing a classic model of dynamic pricing, where consumers buy from one monopolist firm and then another. Then I make two extensions to the model. First, consumers will strategically mask their identity if they have the slightest suspicion their data is being sold. Second, if consumers sell data to the first firm instead of buying a good, everyone will be forced to sell for free.

1 Introduction

In 2018, ProPublica reported how health insurers were buying information about millions of Americans from data brokers (Allen, 2018). They were using data about what TV people watched, what sorts of clothes they bought, and myriad other details to predict medical costs. While insurers are prohibited from using this information to deny people coverage for most plans, there are nevertheless strong incentives to price discriminate. When data brokers trade massive amounts of data, consumers lose track of how data collection can harm them. Many know that their data are being sold. Few know to whom.

This paper explores the consequences of such information asymmetries. I build off a classic model of dynamic pricing, where consumers buy from a (potentially different) monopolist in each of two periods (Taylor, 2004). The first period firm can sell data about who bought their product to the second firm. I review Taylor's results for when consumers are naive (they do not account for the sale of their data) and for when they are sophisticated (consumers do account for data collection). These results are instructive. Privacy is not always good for social welfare. Sophisticated consumers strategically make different purchasing decisions so that the different types of consumers look similar, making data worthless.

I study two extensions to the model. In the previous example with health insurers and data brokers, consumers might suspect that their data were being sold, but they would not be certain about to whom. Suppose, then, that consumers in the model were not fully naive or sophisticated, but believed that the second period firm maybe has their data. (In fact, the second firm always does, but consumers do not know this.) I show that if consumers believed in the slightest that the second firm has their data, they will act as if they were fully sophisticated. This is because the first firm charges consumers their willingness to pay, so consumers are indifferent between buying or not. Then it is costless for some consumers to strategically reject the first good, thereby hiding everyone's types.

My second extension is about markets for data. Consumers rarely can buy goods from digital platforms like YouTube. Rather, they give up their data for a price (in the YouTube case, for free, personalized videos). Suppose the first firm were not a monopolist seller, but a monopsonist buyer of data. If the rest of the game proceeds as before (the second firm is a monopolist who can buy data, and consumers are sophisticated), every consumer will sell their data for zero price. This is because consumers with low valuations of the second good are willing to give up their data for free. Then the only group who do not want to sell data are consumers with high valuations, except now they have revealed themselves by not selling. Hence they also sell at zero price.

There are two more sections in this paper. In the rest of this section, I review selected previous work on the economics of privacy. In Section 2, I review the model by Taylor. In Section 3, I sketch my extensions.

1.1 Previous Literature

In three influential papers, Acquisti, Brandimarte, and Loewenstein (2015), Acquisti, Taylor, and Wagman (2016), and Acquisti (2024) review research on the economics of privacy. They highlight empirical studies which find that consumers do not entirely know what will happen to their data and that consumers experience cognitive biases like myopia.

In contrast, I study such issues theoretically. My setting is that of dynamic pricing, that is, when firms learn to price discriminate against consumers through repeated interactions. There are classic models of dynamic pricing by Taylor (2004) and Acquisti and Varian (2005). I work with Taylor’s simpler model with two periods and naive or sophisticated consumers. Acquisti and Varian (2005)’s alternative model lets consumers use privacy-enhancing technologies like turning off cookies. They show that it is profitable for firms to commit not to use data for pricing, unless there is a large share of myopic (naive) consumers. Continuing this line of research, Ichihashi (2020) studies a monopolist seller of multiple goods. The seller chooses to commit to use consumer data for recommendations but not pricing. In general, data disclosure can provide many benefits and harms, and consumers have multiple countermeasures available. For clarity, I will study the simplest setting.

Most theoretical work on privacy assumes sophisticated consumers. The most popular alternative assumption is that consumers are either fully naive or fully sophisticated. Taylor (2004) assumes all consumers are naive or sophisticated. Acquisti and Varian (2005) studies when some proportion of consumers are naive. Bisceglia, Bonatti, and Scott Morton (2025) make the same assumption, but study a digital platform in one period rather than dynamic pricing. Consumers like cheap services but dislike services which gather data. The firm gathers data about consumers’ valuation of privacy and uses it to tailor the price and quality (privacy) of services offered. When a large enough fraction of consumers are naive, the naive are exploited, giving up data for free. The proportion of naive consumers affects the optimal policy (whether it should involve a cap on total data collected or a mandatory free, private option).

Unlike these previous papers, my work explores when consumers have intermediate beliefs. They might not know anything for sure, but they have some degree of belief as to whether a firm has their data.

An alternative perspective on naive consumers is by Heidhues and Köszegi (2017). For them, firms collect data about whether consumers are naive, rather than data about their willingness to pay. The resulting naivety-based discrimination has very different consequences from price discrimination.

2 Model

This section gives a brief, intuitive explanation of Taylor (2004). I will discuss key results from Taylor (2004), because the same ideas and arguments will come up in my extension. For ease of exposition, I will not be fully rigorous. In general, for this section, numerical examples and diagrams are mine, but other material is directly from Taylor (2004).

2.1 Setup

2.1.1 Consumers and firms

There are two periods $t = 1, 2$. There is measure one of consumers, indexed by $i \in [0, 1]$. In each period $t = 1, 2$, each consumer i demands one unit of good t . Each consumer has an unobserved demand parameter $\lambda_i \in [0, 1]$, which is drawn i.i.d. from some distribution with CDF $F(\lambda)$. The CDF $F(\lambda)$ is commonly known. Consumer i values good t at v_{it} . Their valuation can be either high or low, so $v_{it} \in \{v_L, v_H\}$ with $v_H > v_L > 0$. For each consumer, their valuations v_{i1}, v_{i2} are drawn i.i.d., with $\Pr[v_{it} = v_H | \lambda_i] = \lambda_i$.

There are two firms. Firm t is the monopolist seller of good t .

2.1.2 Timing

The game unfolds as follows. First, each consumer privately observes their valuations for both goods, (v_{i1}, v_{i2}) . Firm 1 then posts a price $p_1 \in \mathbb{R}_+$ and consumers can buy ($q_{i1} = 1$) or reject ($q_{i1} = 0$). Firm 2 observes the offer p_1 .

The game can occur under a “confidential” regime or a “disclosure” regime. Under the “confidential” regime, firm 1 cannot tell firm 2 who bought good 1. Under the “disclosure” regime, firm 1 can offer the customer list to firm 2 for $w \in \mathbb{R}_+$. Mathematically, the customer list is given by $q_{i1} \in \{0, 1\}$ for all $i \in [0, 1]$. Firm 2 may buy ($x = 1$) or reject ($x = 0$) the list.

With the list, firm 2 offers $p_{i2} = p_2^1$ to buyers of good 1 and $p_{i2} = p_2^0$ to those who refuse. Without the list, firm 2 offers $p_{i2} = p_2$ to everyone. Finally, consumers buy ($q_{i2} = 1$) or reject ($q_{i2} = 0$).

2.1.3 Payoffs and solution concept

Both consumers and firms are risk-neutral and do not discount the future.

At the end of the game, consumer i earns payoff $(v_{i1} - p_1)q_{i1} + (v_{i2} - p_{i2})q_{i2}$.

Under the “confidential regime”, firm 1 earns $p_1 Q_1$ and firm 2 earns $p_2 Q_2$, where Q_1, Q_2 are the mass of buyers for goods 1 and 2 respectively. Neither firm has any production costs. The “disclosure regime” payoffs are similar, but firm 2 might pay firm 1 w for data. Concretely, firm 1 earns $p_1 Q_1 + wx$ and firm 2 earns $(1 - x)p_2 Q_2 + x(p_2^1 Q_2^1 + p_2^0 Q_2^0 - w)$, where Q_2^1, Q_2^0 are the masses of consumers who buy at p_2^1, p_2^0 respectively.

The solution concept is efficient Perfect Bayesian Equilibrium (PBE), which is a PBE with social efficiency as a tie-breaking rule.

2.2 The confidential regime

We come to the first result from Taylor (2004).

Proposition 1. *There is a unique efficient PBE under the confidential regime. Equilibrium strategies are*

$$p_t = \begin{cases} v_L, & \text{if } \mathbb{E}[\lambda] \leq v_L/v_H \\ v_H, & \text{if } \mathbb{E}[\lambda] > v_L/v_H \end{cases}$$

$$q_{it} = \begin{cases} 1, & \text{if } p_t \leq v_{it} \\ 0, & \text{if } p_t > v_{it} \end{cases}$$

for any $i \in [0, 1]$ and $t = 1, 2$.

Under the confidential regime, the firms cannot share data, so each faces the same problem. Notice that the only prices the firms will charge are v_L or v_H (in fact this is true under the other regimes too). Intuitively, the firms should charge high prices if demand is inelastic, and low prices if demand is elastic.

Suppose $v_L = 1$ and $v_H = 2$. Then demand is inelastic if more than half the consumers have a high valuation. Formally, $p_1 = p_2 = v_H$ only when $\Pr[v_{it} = v_H] > 1/2$. Notice that

$$\Pr[v_{it} = v_H] = \mathbb{E}[\Pr[v_{it} = v_H | \lambda_i]] = \mathbb{E}[\lambda]$$

so

$$\Pr[v_{it} = v_H] > 1/2 \Leftrightarrow \mathbb{E}[\lambda] > v_L/v_H$$

which is the expression in Proposition 1.

Taylor states his results algebraically. Below I give a graphical interpretation of Proposition 1.

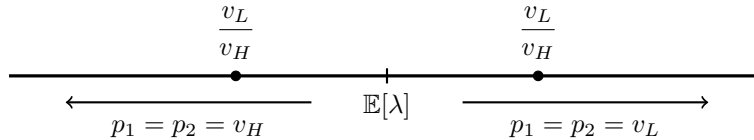


Figure 1: Equilibrium under the confidential regime (Proposition 1)

An important takeaway is that firms set prices by trading off the proportion of high types and the relative sizes of the high and low valuation. Here the proportion of high types is given by the prior probability $\mathbb{E}[\lambda]$.

If firm 2 had data, they could use their posterior $\Pr[v_{i2} = v_H | q_{i1}]$ instead of the prior. Later, we will see that sophisticated consumers can strategically distort their demand q_{i1} . They will be able to set firm 2's posteriors such that firm 2 charges the same price to those who buy and those who refuse good 1.

2.3 Disclosing data with naive consumers

Here our next result, taken from Taylor (2004).

Proposition 2. *There is a unique efficient PBE outcome of the game, as follows.*

1. For some $\bar{\gamma} > 1$, if $\bar{\gamma}\mathbb{E}[\lambda] \leq v_L/v_H$, the prices are $p_1 = p_2^1 = v_L$. The price of the customer list is zero. All customers buy both goods.
2. If $v_L/v_H < \bar{\gamma}\mathbb{E}[\lambda]$, firm 1 charges $p_1 = v_H$. Consumer i buys iff $v_{i1} = v_H$. The price of the customer list is positive. Firm 2 charges $p_2^1 = v_H$ to buyers and $p_2^0 = v_L$ to refusers. Only (v_H, v_L) consumers do not buy good 2.

There is an easier graphical version of this result. This diagram is mine.

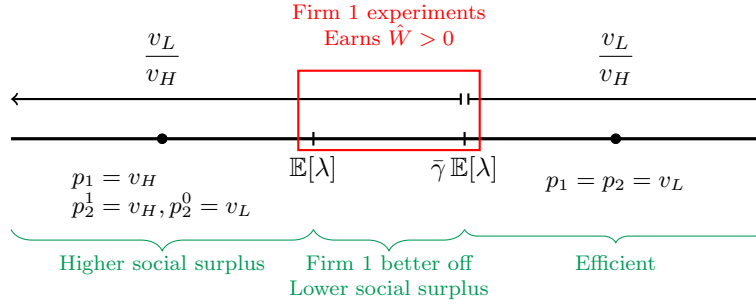


Figure 2: Black text describes the equilibrium (Proposition 2).
Green text indicates welfare (Corollary 1 in Taylor (2004)).

To the right of $\bar{\gamma}\mathbb{E}[\lambda]$, we have the same efficient low prices as before. Between $\mathbb{E}[\lambda]$ and $\bar{\gamma}\mathbb{E}[\lambda]$, firm 1 charges v_H when it would have charged v_L before. This is because those who buy at v_H reveal themselves to be high types. In this region, when demand is moderately elastic, firm 1 can make more money by selling the customer list. To the left of $\mathbb{E}[\lambda]$, demand is inelastic, so firm 1 charges a high price and firm 2 price discriminates.

Surprisingly, data disclosure is not necessarily good for bad for social surplus. When demand is moderately elastic, data disclosure is worse compared to the confidential regime because firm 1 raises prices to collect more data. When demand is inelastic, price discrimination improves welfare because it makes the good more affordable to those who would otherwise have been excluded from purchasing.

One short note is in order. I believe Taylor is abusing terminology by saying this is a PBE. Technically, the consumer beliefs are not consistent with the equilibrium strategies. When $\mathbb{E}[\lambda] < v_L/v_H \leq \bar{\gamma}\mathbb{E}[\lambda]$ consumers know firm 1 should set $p_1 = v_L$. But firm 1 sets $p_1 = v_H$, so the consumers should realize it is up to no good (by selling their data)!

2.4 Disclosing data with sophisticated consumers

We finally arrive at Taylor's main results. I will keep his numbering.

Lemma A4. *Consider the continuation game that begins after firm 1 sets p_1 .*

If $p_1 = v_L$, then there exists a PBE of the continuation game in which all consumers purchase good 1.

If $p_1 = v_H$, then there exists a PBE of the continuation game where all type- (v_L, v_L) and (v_L, v_H) consumers refuse to purchase good 1. All (v_H, v_L) buy, and a fraction ρ^ of (v_H, v_H) consumers refuse.*

No other PBE of the respective continuation games yields a higher expected payoff to firm 1.

This result is a little involved to prove. Intuitively, some high types strategically refuse to buy good 1 and pretend to be low types. Critically, the (v_H, v_H) consumers have the power to make the customer list worthless!

Let me illustrate this with a numerical example. Suppose there were 4 (v_L, v_L) types and 6 (v_H, v_H) types. The following would not be an equilibrium: 4 (v_L, v_L) , 3 (v_H, v_H) refuse good 1, and 3 (v_H, v_H) buy. Firm 2 would charge $p_2^0 = v_L$ and $p_2^1 = v_H$, so (v_H, v_H) would want to deviate (by refusing to buy good 1). Instead, the equilibrium would be as follows: 4 (v_L, v_L) , 4 (v_H, v_H) refuse good 1 and 2 (v_H, v_H) buy. Firm 2 would charge $p_2^0 = p_2^1 = v_H$.

The general principle is that ρ^* of (v_H, v_H) refuse, and ρ^* calibrated to force firm 2 to charge the same prices to buyers and refusers.

Proposition 4. *There is a unique efficient PBE outcome as follows.*

1. If $\mathbb{E}[\lambda] \leq v_L/v_H$, then firm 1 charges $p_1 = v_L$. The price of the customer list is $\hat{W} = 0$, and firm 2 charges $p_2 = v_L$. All consumers purchase good 1 and 2.
2. For some $\underline{\gamma} < 1$, if $\underline{\gamma}\mathbb{E}[\lambda] \leq v_L/v_H < \mathbb{E}[\lambda]$, then firm 1 charges $p_1 = v_L$ and all consumers buy. The price of the customer list is $\hat{W} = 0$ and firm 2 sets $p_2^1 = v_H$. Consumer i buys iff $v_{i2} = v_H$.
3. If $v_L/v_H < \underline{\gamma}\mathbb{E}[\lambda]$, firm 1 charges $p_1 = v_H$. All (v_H, v_L) and $(1 - \rho^*)$ of (v_H, v_H) consumers buy. The price of the customer list is zero and firm 2 sets $p_2^0 = p_2^1 = v_H$. Consumer i buys iff $v_{i2} = v_H$.

Taylor's results are easier graphically. The diagram below is mine.

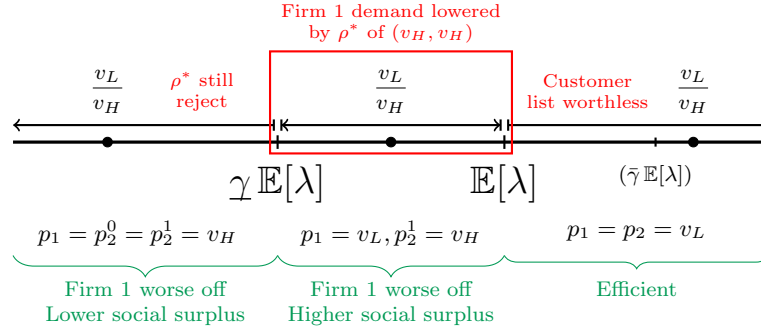


Figure 3: Black text explains the equilibrium (Proposition 4).
 Green text indicates welfare (Corollary 2 in Taylor (2004)).

Demand is elastic to the right of $\mathbb{E}[\lambda]$, so we have efficient low prices. Because sophisticated consumers make the customer list worthless, firm 1 no longer has an incentive to raise prices to experiment between $\mathbb{E}[\lambda]$ and $\bar{\gamma}\mathbb{E}[\lambda]$. In the region between $\underline{\gamma}\mathbb{E}[\lambda]$ and $\mathbb{E}[\lambda]$, demand is moderately inelastic. Firm 2 finds it profitable to set high prices, but firm 1 cannot. This is because if firm 1 tried to raise prices, ρ^* of the (v_H, v_H) types would strategically refuse to buy. To the left of $\underline{\gamma}\mathbb{E}[\lambda]$, demand is very inelastic, and both firms set high prices.

Again, data disclosure is not necessarily better or worse than the confidential benchmark. When demand is moderately inelastic, strategic high types can keep the first period price low, improving welfare. When demand is very inelastic, the same high prices obtain, but compared with the confidential regime, the high types' refusal to buy good 1 reduces social surplus.

Notice also that firm 1 is always worse off than in the confidential regime. A typical result in this literature is that firm 1 would like to commit to privacy.

3 Extensions

Reading Taylor (2004) is edifying in its own right. Another benefit is that now I have all the necessary tools to make my long-promised extensions.

3.1 Intermediate beliefs

Definition 1. (Intermediate belief θ) All consumers believe firm 2 is sold the customer list with probability θ .

Remark. Without loss of generality, firm 1 sells the customer list to firm 2 with probability 1.

Here is my main result. Surprisingly, even very uncertain consumers protect their identity as if they were perfectly sophisticated.

Claim 1. For $\theta \in (0, 1)$, the strategies and beliefs from Proposition 4 are still a PBE. In particular, if $p_1 = v_H$, then ρ^* of (v_H, v_H) strategically refuse good 1.

This is because high types refuse to buy good 1 to hide their identity. Such refusals are costless because the monopolist firm charges exactly their willingness to pay v_H .

Proof. (Sketch)

If $p_1 = v_L$, everyone buys good 1. If $p_1 = v_H$, types (v_L, v_L) and (v_L, v_H) refuse regardless of θ , and (v_H, v_L) always buys. Only the behaviour of (v_H, v_H) remains to be characterized.

If they believe firm 2 is buying their data, ρ^* will reject good 1. If they believe their data is kept private, then at $p_1 = v_H$, they are indifferent between buying and not. Hence, regardless of their beliefs, ρ^* of (v_H, v_H) rejecting is optimal.

Firm beliefs and best responses are as before. The firms are still trading data, and I just proved the consumers take the same strategy. Therefore we have the PBE from Proposition 4 for all $\theta \in (0, 1)$. $\square$

This surprising result can be understood like so: if one can costlessly protect their privacy, and one is worried about data collection, then there is no reason not to do so. However, it seems unrealistic that privacy protection is costless. For one, perhaps real-world firms do not exactly charge consumers' willingness to pay.

3.2 Data markets

Definition 2. (Market for data) The market for data operates as follows. As in the original game, consumers have valuations (v_{i1}, v_{i2}) . They offer to sell v_{i1} to firm 1 for p_{i1} . The first firm is a single buyer of data. It does not sell any goods. The rest of the game proceeds similarly to the original game under the disclosure regime with sophisticated customers. Firm 1 may sell the list of v_{i1} that it bought to firm 2. Firm 2 can use this data to price discriminate.

Claim 2. In equilibrium, all consumers sell v_{i1} for the same price of zero.

Proof. (Very rough sketch.)

Type (v_L, v_L) is willing to sell their data for any $p_{i1} \geq 0$. Because firm 1 is a monopsonist, it sets the price as low as possible, at $p_{i1} = 0$. To blend in with (v_L, v_L) , type (v_L, v_H) will also sell their data for the same price of zero.

Now if $(v_H, \cdot)$ don't sell, they will reveal themselves as high types. Being sophisticated, (v_H, v_L) and (v_H, v_H) realize they may as well sell their data anyway. They can only sell at price zero. Thus, for all consumers i , $p_{i1} = 0$.

I omit the rest of the argument for why this is a PBE, but clearly firm 2 will charge $p_{i2} = v_L$ if $v_{i1} = v_L$ and $p_{i2} = v_H$ if $v_{i1} = v_H$ (or if a consumer refuses to sell their data). $\square$

In Claim 1, we saw that if the private option is costless, it will have perfect take-up. One can roughly understand Claim 2 to say that when the private option is costly, it will have zero take-up. This is because only the high types would pay for the private option, but then they would be giving themselves away, so they must not be willing to pay for privacy after all.

This result seems counterintuitive. Many people go out of their way to protect their privacy. The result makes more sense when we focus on price discrimination. Returning to our initial health insurance context, perhaps non-smokers would be happy to disclose their lifestyle habits, and only smokers would want to hide. In further work, one could consider extending this model to have more than two types, or to introduce goods of different levels of quality, and look for when consumers start paying for privacy.

Let me state one final result. The idea is similar to Claim 1.

Claim 3. *Even if consumers have some intermediate belief $\theta \in [0, 1]$ that firm 2 were buying their data, everyone will still sell their type for free.*

Proof. (Sketch) If consumers believed firm 2 would not get their data, they would sell at any $p_{i1} \geq 0$. If consumers believed firm 2 would get their data, they would also sell by Claim 2. Hence they behave the same, no matter what they believe. Firm responses are unchanged from Claim 2 because they are still secretly trading data. $\square$

4 Conclusion

I derive sharp results for privacy when consumers do not know who has their data, and for when consumers sell their data directly to a platform. I suspect these knife-edge results come about because my model is so simple. Next steps include formalizing the proofs of Claims 2 and 3 and finding a richer setting producing different, more realistic behaviour.

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